Horn kernels and normalization of diagonal simplicial tensor modules

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Abstract

Deleting or repeating the same index in every axis of a tensor gives concrete examples of simplicial modules over a nonzero commutative ring. We use these examples to study horn filling, normalization, chain contractions, and homology through coordinate calculations, supported by a Python implementation. Horn kernels have explicit bases, and fillers are unique precisely in dimensions above the number of tensor axes. The normalized complex splits into two-term complexes with identity differential, giving explicit equivariant contractions. Two shapes give isomorphic simplicial modules exactly when they differ by a permutation and addition of the same integer to every entry. Although the full module is contractible, a submodule generated by a single matrix can have nonzero first homology. We describe generated submodules as marked quotients. An appendix studies their kernel families over an infinite field, including an explicit parameterization for 3×3 matrices.

1 Introduction

Matrices and higher-order arrays provide a computational way to study simplicial modules. Deleting or repeating the same index in every axis defines face and degeneracy operators; allowing the dimensions to vary together gives a simplicial module. The *SimplicialTensors* repository [5] implements these operations, the alternating boundary, and horn filling in Python. This paper develops the tensor example as a concrete route through the corresponding constructions of simplicial algebra.

For instance, horn equations ask whether a compatible collection of diagonal faces determines a tensor and, when it does not, how its fillers differ. Normalization removes the chains obtained by degeneracy operators. A chain contraction then supplies an explicit filling of each cycle in the full tensor module. These constructions can be examined both on coordinate tensors and in array computations.

These questions admit coordinate answers. A face kills a coordinate tensor precisely when its index tuple contains the deleted index. Consequently a horn kernel has a basis indexed by tuples covering the prescribed face indices. Its rank follows by inclusion–exclusion, and horn fillers are unique exactly above the number of tensor axes (Theorem 3.2). The same basis splits the normalized complex into disks (Proposition 4.1), yielding explicit contractions. Two shapes give isomorphic simplicial modules exactly when they differ by a permutation and addition of the same integer to every entry.

The ambient simplicial module is contractible, but a submodule generated by one tensor need not be. A 2×2 matrix gives an elementary example with one-dimensional first homology (Example 2.1). We describe a generated submodule as a marked quotient of the free simplicial module on a simplex. Normalization makes its kernel finite-dimensional data (Section 6). Appendix A proves

that the closure of the generic kernel image is rational over an infinite field and computes the image and quotient homology for shape $(3,3)$. This appendix additionally uses Grassmannians, incidence schemes, and function fields.

We use the normalization and Dold–Kan theorems [1, 4, 10]. The horn-space isomorphism of [7, Lemma 2.10], stated there for simplicial objects in finite-dimensional real vector spaces, follows here from a horn-complement argument over A. All coordinate calculations are valid over a nonzero commutative ring; Section 6 works over a field.

2 The construction

Throughout, A is a nonzero commutative ring with identity, $k \geq 1$, and $[p] = \{0, \dots, p\}$. Let Δ denote the category of finite nonempty ordinals and nondecreasing maps. Fix a shape $\vec{s} = (n_1, \dots, n_k) \in \mathbb{Z}_{>0}^k$ and put

$$n = \min_a n_a - 1, \qquad c_a = n_a - 1 - n, \qquad I_p = \prod_{a=1}^{k} [p + c_a].$$

The integers $c_a = n_a - \min_b n_b$ are the *shape offsets*. Thus $c_a \geq 0$ and at least one c_a is zero. The integer n records the degree in which tensors of the original shape occur; the offsets (c_a) determine the entire simplicial module.

For $c \geq 0$, put $J_c([p]) = [p + c]$. For $\alpha : [p] \to [q]$, define

$$J_c(\alpha)(r) = \begin{cases} \alpha(r), & 0 \leq r \leq p, \\ q + t, & r = p + t, \quad 1 \leq t \leq c. \end{cases}$$

This defines a covariant functor $J_c : \Delta \to \mathbf{Set}$, since composition acts separately on the initial ordinal and the final c elements. Consequently $I = \prod_a J_{c_a}$ is cosimplicial and

$$X_p = X_p(\vec{s}; A) = A^{I_p}, \qquad X(\alpha)(T) = T \circ I(\alpha)$$

defines a functor $X : \Delta^{\mathrm{op}} \to \mathbf{Mod}_A$. We call X the *diagonal simplicial tensor module* of shape $\vec{s}$. Write E_m for the indicator function of $m \in I_p$.

For $p \geq 1$ and $0 \leq i \leq p$, the index coface $\Delta_i : I_{p-1} \to I_p$ inserts a gap at i in every coordinate; its image consists of the tuples avoiding i. For $p \geq 0$ and $0 \leq i \leq p$, the codegeneracy $\Sigma_i : I_{p+1} \to I_p$ identifies i and $i + 1$ in every coordinate. Thus $d_i T = T \circ \Delta_i$ and $s_i T = T \circ \Sigma_i$. In particular,

$$d_i E_m = \begin{cases} 0, & i \in \operatorname{im}(m), \\ E_{\Delta_i^{-1} m}, & i \notin \operatorname{im}(m), \end{cases} \qquad \operatorname{im}(m) = \{m_1, \dots, m_k\}. \tag{1}$$

The nonzero images in this formula are distinct basis elements. There are no outgoing face maps in degree zero. The associated chain complex has $X_{-1} = 0$ and differential

$$\partial_p = \sum_{i=0}^{p} (-1)^i d_i \quad (p \geq 1), \qquad \partial_0 = 0.$$

For any simplicial module Y, we write $H_p(Y) = \ker \partial_p / \operatorname{im} \partial_{p+1}$. When $k = 1$, $X_p = A^{p+1}$; a face of a two-entry vector is a one-entry vector.

In the repository's `simplicial_tensors.tensor_ops` module, `face(T,i)` and `degen(T,i)` compute these face and degeneracy maps, and `bdry(T)` computes the alternating face boundary in positive degrees. In degree zero, the empty array returned by `bdry(T)` represents the zero boundary. Install with `pip install -e .`, then run `python experiments/dstm_examples.py` to reproduce Examples 2.1 and 3.3. The script uses integer arrays and exact ranks over $\mathbb{Q}$: it reports first homology dimension 1 for the generated matrix example and 2 free parameters for the horn. These calculations illustrate the examples; they do not implement a general homology algorithm.

The construction is the diagonal of the k-fold simplicial module $A^{\prod_a [p_a+c_a]}$; compare the diagonal construction in [6]. Equivalently, $X \cong \bigotimes_{a=1}^{k} A^{J_{c_a}}$, with degreewise tensor product. All structure maps are defined over $\mathbb{Z}$, and coordinate bases give

$$X(\vec{s}; A) \otimes_A B \cong X(\vec{s}; B) \tag{2}$$

for every homomorphism $A \to B$.

For $T \in X_n$, its generated simplicial submodule has degree-p part

$$\langle T \rangle_p = \operatorname{span}_A \{X(\alpha)(T) : \alpha : [p] \to [n]\}.$$

This is closed under faces and degeneracies by functoriality.

Example 2.1. Over a field K, take shape $(2, 2)$, write $E_{ij} = E_{(i,j)}$ and let

$$T = E_{01} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad Y = \langle T \rangle \subseteq X.$$

Both faces vanish: $d_0 T = d_1 T = 0$. Consequently $Y_0 = 0$, $Y_1 = KT$, and

$$Y_2 = \operatorname{span}_K \{s_0 T, s_1 T\}, \qquad s_0 T = E_{02} + E_{12}, \quad s_1 T = E_{01} + E_{02}.$$

The simplicial identities give

$$\partial_2 s_0 T = T - T + s_0 d_1 T = 0, \qquad \partial_2 s_1 T = s_0 d_0 T - T + T = 0.$$

Thus T is a nonzero cycle that is not a boundary in Y, and $H_1(Y) \cong K$. Every simplex of Y above degree 1 is a sum of degeneracies.

In the ambient complex, however,

$$\partial_2 E_{12} = T.$$

This primitive does not belong to Y_2. Indeed, expressing E_{12} as $a(E_{02} + E_{12}) + b(E_{01} + E_{02})$ would force $b = 0$, $a + b = 0$, and $a = 1$, a contradiction. Thus a cycle in a simplicial submodule can bound in the ambient module without bounding in the submodule.

3 Horn kernels and normalization

For any simplicial A-module Y, put

$$N_p(Y) = \bigcap_{i=1}^{p} \ker d_i, \qquad D_p(Y) = \sum_{i=0}^{p-1} \operatorname{im} s_i.$$

Here $N_0(Y) = Y_0$, $D_0(Y) = 0$, and the normalized differential is d_0 in positive degrees and zero in degree zero; this is the convention in [10, Exercise 8.3.4]. The normalization theorem gives

$Y_p = N_p(Y) \oplus D_p(Y)$ [1, Corollary 1.12]. The Dold–Kan theorem gives an equivalence with non-negative chain complexes [10, Theorem 8.4.1].

For $p \geq 1$ and $0 \leq j \leq p$, let

$$R_{p,j}(Y) = \bigcap_{i \neq j} \ker d_i.$$

The module $\Lambda_{p,j}(Y)$ of compatible horns consists of tuples $(y_i)_{i \neq j}$ with $y_i \in Y_{p-1}$ satisfying $d_i y_\ell = d_{\ell-1} y_i$ for $i < \ell$, $i, \ell \neq j$; for $p = 1$ there are no compatibility conditions. The horn map is $\Phi_j : Y_p \to \Lambda_{p,j}(Y)$, $y \mapsto (d_i y)_{i \neq j}$. Every simplicial module is Kan, so Φ_j is surjective [10, Lemma 8.2.8]. Its fibers are torsors for $R_{p,j}(Y)$.

Lemma 3.1. *For every simplicial A-module Y, $p \geq 1$ and $0 \leq j \leq p$,*

$$Y_p = R_{p,j}(Y) \oplus D_p(Y).$$

The restriction $\Phi_j|_{D_p(Y)}$ is an isomorphism. Thus every horn has a unique filler in the submodule $D_p(Y)$.

Proof. For $0 \leq j < p$, the simplicial identities give inverse maps

$$F_j = 1 - s_j d_j : R_{p,j}(Y) \longrightarrow R_{p,j+1}(Y), \qquad G_j = 1 - s_j d_{j+1} : R_{p,j+1}(Y) \longrightarrow R_{p,j}(Y).$$

Indeed, the required faces vanish by $d_i d_j = d_{j-1} d_i$ for $i < j$ and $d_j s_j = d_{j+1} s_j = 1$. For $x \in R_{p,j}$, $d_{j+1} F_j x = -d_j x$, whence $G_j F_j x = x$; the other composition is identical with j and $j+1$ interchanged. Both maps induce the identity modulo $D_p(Y)$. Since $R_{p,0}(Y) = N_p(Y)$ maps isomorphically onto $Y_p / D_p(Y)$, the same holds for every $R_{p,j}(Y)$. This proves the direct sum. The final assertion follows from surjectivity of Φ_j and $\ker \Phi_j = R_{p,j}(Y)$. □

For simplicial objects in finite-dimensional real vector spaces the horn-space isomorphism is also proved in [7, Lemma 2.10]; Lemma 3.1 gives an argument over arbitrary A.

Theorem 3.2. *For the diagonal simplicial tensor module X, let $F_j = [p] \setminus \{j\}$. Then*

$$R_{p,j}(X) = \bigoplus_{\substack{m \in I_p \\ F_j \subseteq \mathrm{im}(m)}} A E_m, \qquad \operatorname{rank} R_{p,j}(X) = \sum_{t=0}^{p} (-1)^t \binom{p}{t} \prod_{a=1}^{k} (p + c_a + 1 - t). \tag{3}$$

In particular, $R_{p,j}(X) \neq 0$ if and only if $p \leq k$. Thus horn fillers are nonunique for $1 \leq p \leq k$ and unique for $p > k$, independently of j and of the offsets.

Proof. By (1), $d_i T = 0$ exactly when T is supported on tuples containing i. Intersecting these support conditions gives (3). Inclusion–exclusion counts tuples containing all p elements of F_j: forbidding a specified t of these elements leaves $p + c_a + 1 - t$ choices in coordinate a. A tuple of length k cannot contain $p > k$ distinct values. If $p \leq k$, assign the p required values to distinct coordinates and fill the remaining coordinates arbitrarily in $[p]$. This gives a basis element, nonzero because $A \neq 0$. □

Example 3.3. For shape $(3, 3)$, consider the 2-horn with missing face 0 and prescribed faces

$$d_1 T = \begin{pmatrix} 1 & 3 \\ 7 & 9 \end{pmatrix}, \qquad d_2 T = \begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}.$$

The compatibility condition $d_1d_2T = d_1d_1T$ holds: both sides are the one-entry matrix (1). Every filler is of the form

$$T = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & u \\ 7 & v & 9 \end{pmatrix}, \qquad u, v \in A.$$

Thus the difference of two fillers lies in $R_{2,0}(X) = AE_{(1,2)} \oplus AE_{(2,1)}$, whose basis consists of the two index pairs containing both 1 and 2.

An algebraic r-hypergroupoid is a simplicial module whose horns have unique fillers in every dimension greater than r [2, 3]. Theorem 3.2 says that X is such an object exactly when $r \geq k$; its least possible value of r is k. In particular, for the original tensor degree $n = \min_a n_a - 1$, the n-hypergroupoid condition is $k \leq n$, with nonunique n-horn fillers when $k = n \geq 1$.

4 The normalized complex

Set $Z_p = \ker(d_0 : N_p(X) \to N_{p-1}(X))$ for $p \geq 1$ and $Z_0 = X_0$. Since some $c_a = 0$, every index in I_0 contains zero. Therefore, in all degrees $p \geq 0$,

$$Z_p = \bigoplus_{\substack{m \in I_p \\ [p] \subseteq \mathrm{im}(m)}} AE_m. \tag{4}$$

Proposition 4.1. *For $p \geq 1$ and $0 \leq j \leq p$, define $\iota_j : Z_{p-1} \to R_{p,j}(X)$ by $\iota_j(E_m) = E_{\Delta_j m}$. Then*

$$R_{p,j}(X) = Z_p \oplus \iota_j Z_{p-1}, \qquad d_j \iota_j = 1.$$

Consequently, writing $D^r(M)$ for the chain complex $M \xrightarrow{1} M$ in degrees $r, r-1$, there is a coordinate-defined isomorphism

$$N_\bullet(X) \cong \bigoplus_{q=0}^{k-1} D^{q+1}(Z_q). \tag{5}$$

In particular $N_\bullet(X)$ is split exact.

Proof. Partition the basis in (3) according as the tuple contains j or avoids j. The first part is precisely the basis of Z_p. Deleting the gap at j identifies the second part bijectively with the basis of Z_{p-1}, and (1) describes d_j on both parts. Taking $j = 0$ pairs each basis element of Z_q with its coordinatewise shift by 1 in degree $q + 1$. There are no other basis elements of $N_\bullet(X)$. Finally $Z_q = 0$ for $q \geq k$, because k coordinates cannot contain $q + 1$ distinct values. □

Explicitly, the contraction $h_p : N_p(X) \to N_{p+1}(X)$ is

$$h_p(E_m) = \begin{cases} E_{m+\mathbf{1}}, & 0 \in \mathrm{im}(m), \\ 0, & 0 \notin \mathrm{im}(m). \end{cases}$$

Thus $d_0h + hd_0 = 1$, with the differential interpreted as zero in degree zero. The bases in (3) and (4) also show that horn kernels, normalized chains and normalized cycles commute with arbitrary base change.

Corollary 4.2. *Put $b_p = \operatorname{rank} Z_p$ and $b_{-1} = 0$. Then*

$$b_p = \sum_{t=0}^{p+1} (-1)^t \binom{p+1}{t} \prod_{a=1}^{k} (p + c_a + 1 - t), \qquad \operatorname{rank} N_p(X) = b_p + b_{p-1}.$$

For constant shapes, $c_a = 0$ for every a, and

$$b_p = (p+1)! \left\{ {k \atop p+1} \right\}, \qquad \operatorname{rank} N_p(X) = p! \left\{ {k \atop p} \right\} + (p+1)! \left\{ {k \atop p+1} \right\},$$

where $\left\{ {k \atop r} \right\}$ is a Stirling number of the second kind.

Proof. Apply inclusion–exclusion to (4) and then use Proposition 4.1. In the constant-shape case the indices covering $[p]$ are exactly the surjections from a k-element set onto a $(p+1)$-element set, counted by $(p+1)! \left\{ {k \atop p+1} \right\}$ [8, §1.4]. □

Example 4.3. For $k = 2$ and any constant shape (N, N), one has $X_p = A^{(p+1)\times(p+1)}$ and

$$N_\bullet(X) \cong D^1(A) \oplus D^2(A^2).$$

The normalized ranks are $1, 3, 2$ in degrees $0, 1, 2$ and zero thereafter. In degree 2, the two basis elements of $R_{2,0}$ are $E_{(1,2)}$ and $E_{(2,1)}$. They map under d_0 to $E_{(0,1)}$ and $E_{(1,0)}$, respectively.

Corollary 4.4. *Over a fixed nonzero commutative ring A, two diagonal simplicial tensor modules are isomorphic if and only if one shape is obtained from the other by permuting its entries and adding the same integer to every entry.*

Proof. An isomorphism gives equality of the ranks of the finite free degree-p modules for every $p \geq 0$. Hence their integer polynomials $\prod_a (x + c_a + 1)$ agree at all nonnegative integers, and therefore agree identically. Their roots, with multiplicities, show that each offset occurs equally often in both shapes. This is equivalent to the stated relation between the shapes. Conversely, a permutation matching the offsets induces a simplicial isomorphism by permuting coordinates. □

Adding the same integer to every shape entry changes the degree in which the original tensor occurs, while preserving the simplicial module.

5 Equivariant contraction

Let $G = \operatorname{Stab}(\vec{s}) = \{\sigma \in S_k : n_{\sigma(a)} = n_a \text{ for all } a\}$. It acts on X_p by permuting tensor axes: $\sigma E_m = E_{\sigma m}$, where $(\sigma m)_a = m_{\sigma^{-1}(a)}$. Since the index maps apply the same operation in every coordinate, this action commutes with all faces and degeneracies. In particular, it acts on the chain complex $(X_\bullet, \partial)$. A *G-equivariant chain contraction* is a family of A-linear maps $H_p : X_p \to X_{p+1}$ such that

$$\partial_{p+1} H_p + H_{p-1} \partial_p = 1_{X_p}, \qquad H_p \sigma = \sigma H_p,$$

where $X_{-1} = 0$ and $H_{-1} = 0$.

Proposition 5.1. *The maps $H_p(E_m) = E_{m+\mathbf{1}}$ define a G-equivariant chain contraction of $X_\bullet$.*

Proof. The shift is well-defined because $m_a \le p + c_a$ implies $m_a + 1 \le p + 1 + c_a$. No coordinate of $m + \mathbf{1}$ is zero, so deleting the gap at zero gives $d_0 H_p(E_m) = E_m$.

For $p \ge 1$ and $1 \le i \le p+1$, we claim

$$d_i H_p = H_{p-1} d_{i-1}. \tag{6}$$

If $i - 1 \in \mathrm{im}(m)$, then $i \in \mathrm{im}(m + \mathbf{1})$; both sides vanish on E_m by (1). Otherwise, both faces survive, and the coordinate gap maps satisfy

$$\Delta_i^{-1}(m + \mathbf{1}) = \Delta_{i-1}^{-1}(m) + \mathbf{1}.$$

Indeed, deleting i subtracts one from a coordinate greater than i, and leaves a coordinate less than i unchanged. Applying this rule before or after the shift gives the displayed identity, proving (6). Hence, for $p \ge 1$,

$$\partial_{p+1} H_p = 1 + \sum_{i=1}^{p+1} (-1)^i H_{p-1} d_{i-1} = 1 - H_{p-1} \partial_p.$$

In degree zero, some c_a is zero, so every $m \in I_0$ contains zero. Thus $m + \mathbf{1}$ contains 1, giving $d_1 H_0 = 0$ and $\partial_1 H_0 = (d_0 - d_1) H_0 = 1$.

Finally, coordinate permutations commute with the shift: $\sigma(m + \mathbf{1}) = \sigma m + \mathbf{1}$. Therefore $\sigma H_p(E_m) = E_{\sigma m + \mathbf{1}} = H_p \sigma(E_m)$. □

For constant shapes, $G = S_k$. Define the symmetric submodule by $\sigma T = T$ for every $\sigma \in S_k$, and the alternating submodule by $\sigma T = \mathrm{sgn}(\sigma) T$. The structure maps preserve these submodules because they commute with G. The contraction preserves them as well: if $\sigma T = \chi(\sigma) T$, where χ is the trivial or sign character, then

$$\sigma H_p(T) = H_p(\sigma T) = \chi(\sigma) H_p(T).$$

Restricting the contraction identity therefore contracts both chain subcomplexes over any A; no averaging or division by $k!$ is required. The normalized contraction h is separately equivariant: its defining condition $0 \in \mathrm{im}(m)$ is invariant under coordinate permutations, and its nonzero values use the same coordinatewise shift.

Over $\mathbb{C}$, complex conjugation commutes with the structure maps and with H and h, since their coordinate formulas have real coefficients. For $k = 2$ and a square shape, the axis transposition is matrix transpose in each degree. Consequently conjugate transpose $T \mapsto T^* = \overline{T}^{\mathsf{T}}$ acts on the underlying real complexes $X_\bullet$ and $N_\bullet(X)$ and commutes with H and h, respectively. Its $+1$ and -1 eigenspaces, the Hermitian and skew-Hermitian matrices, are real chain subcomplexes on which the respective contractions restrict.

Remark 5.2. The decreasing filtration

$$\mathcal{F}^t X_p = \mathrm{span}_A\{E_m : \min_a m_a \ge p + t\}, \qquad t \in \mathbb{Z},$$

is preserved by ∂ and H. For a surviving face, $\min_a(\Delta_i^{-1} m)_a \ge \min_a m_a - 1$, which gives $d_i \mathcal{F}^t X_p \subseteq \mathcal{F}^t X_{p-1}$. The shift raises the minimum by exactly one, so $H_p \mathcal{F}^t X_p \subseteq \mathcal{F}^t X_{p+1}$. The filtration is finite in each degree: $\mathcal{F}^{-p} X_p = X_p$ and $\mathcal{F}^1 X_p = 0$, since $\min_a m_a \le p$. Thus the contraction identity descends to the associated graded complex. Its homology, the E_1 page of the associated spectral sequence, is zero.

6 Generated submodules

Throughout this section, K is a field and $X = X(\vec{s}; K)$. We describe the submodule $\langle T \rangle$ from Section 2 by its normalized kernel. This gives finite-dimensional data for computing its homology.

Let $P = K[\Delta^n]$ be the free simplicial K-module on the standard n-simplex, so P_p has basis the nondecreasing maps $[p] \to [n]$. For $T \in X_n$, its representing property gives

$$f_T : P_\bullet \longrightarrow X_\bullet, \qquad f_T(\alpha) = X(\alpha)(T) \quad \big(\alpha : [p] \to [n]\big).$$

Its image is the simplicial submodule $\langle T \rangle$ generated by T, and $P/\ker f_T \cong \langle T \rangle$. Put

$$C = N(P), \qquad g_T = N(f_T), \qquad L(T) = \ker g_T.$$

Normalization is exact, so

$$N(\langle T \rangle) \cong C/L(T). \tag{7}$$

Here C_p has dimension $\binom{n+1}{p+1}$ for $0 \le p \le n$ and is zero otherwise. Hence the finitely many spaces $L_p(T)$ determine the entire simplicial kernel by Dold–Kan. They classify the quotient with its distinguished generating n-simplex; they need not determine its inclusion in X, and different kernels can yield isomorphic simplicial modules after the generator is forgotten.

Homology is computed from normalized complexes. Since $H_0(C) = K$ and $H_j(C) = 0$ for $j > 0$, the short exact sequence $0 \to L(T) \to C \to C/L(T) \to 0$ gives

$$\begin{aligned} H_j(\langle T \rangle) &\cong H_{j-1}(L(T)) \qquad (j \ge 2), \\ 0 \longrightarrow H_1(\langle T \rangle) \longrightarrow H_0(L(T)) &\xrightarrow{\epsilon} K \longrightarrow H_0(\langle T \rangle) \longrightarrow 0, \end{aligned} \tag{8}$$

where ϵ is induced by the augmentation sending every vertex of Δ^n to 1.

For the matrix in Example 2.1, $N(\langle E_{01} \rangle)$ is K in degree 1 and zero elsewhere: all its higher simplices are degenerate.

A Kernel families over an infinite field

Assume now that K is infinite. Retain the notation P, C, g_T, and $L(T)$ from Section 6. We parameterize the kernels on the locus where their dimensions are minimal.

For a finite-dimensional K-vector space V and $0 \le r \le \dim V$, write $\mathrm{Gr}(r, V)$ for the Grassmannian of r-dimensional linear subspaces of V. This is a projective K-variety whose points over a field extension F/K are the r-dimensional F-linear subspaces of $V \otimes_K F$. Its tautological subbundle of the trivial bundle with fiber V has fiber L at the point representing $L \subseteq V$.

View $W = X_n$ as an affine space. For $0 \le p \le n$, let r_p be the maximal rank of $g_{T,p}$ and put $\ell_p = \dim C_p - r_p$. The locus $U \subset W$ on which all these ranks are attained is a nonempty open subset. Constant-rank kernels define a morphism

$$\Psi : U \longrightarrow G = \prod_{p=0}^{n} \mathrm{Gr}(\ell_p, C_p), \qquad T \longmapsto (L_p(T))_p. \tag{9}$$

Its values satisfy $d_0 L_p \subseteq L_{p-1}$, so they are chain subcomplexes of C. By Chevalley's theorem [9, Tag 054K], the image $\mathcal{M} = \Psi(U)$ is constructible. Give its Zariski closure $Y = \overline{\mathcal{M}}$ the reduced structure. When $\mathcal{M}$ is locally closed, it too is given the reduced induced structure. Thus $\mathcal{M}$ records marked quotients of P, rather than isomorphism classes of unmarked simplicial modules. A K-variety is *rational over* K if it is birational over K to an affine space $\mathbb{A}^d_K$.

Proposition A.1. *The variety $\overline{\mathcal{M}}$ is rational over K.*

Proof. The closure is integral because U is irreducible. Each $g_{T,p}$ depends linearly on T. For a tuple $L = (L_p)$, the conditions

$$g_{T,p}(L_p) = 0 \quad (0 \le p \le n)$$

therefore define a linear subspace $V_L \subset W$. Let $\mathcal{E} \subset W \times Y$ be the incidence scheme cut out by these conditions on the tautological subbundles L_p pulled back from the factors of G. Over $U \times Y$, L_p and $\ker g_{T,p}$ are subbundles of equal rank, so inclusion forces equality. Hence $\mathcal{E} \cap (U \times Y)$ is the graph of Ψ as a scheme, and its fiber over L is $U \cap V_L$.

Let $F = K(Y)$ be the function field of Y, let η be its generic point, and write W_F, U_F for base change from K to F. Let $V_\eta \subset W_F$ be the linear subspace defined by the incidence conditions at η. Its intersection with U_F is nonempty, since Ψ is dominant onto its image closure. Put $d = \dim_F V_\eta$. Choose d coordinate functions on W whose restriction gives an isomorphism $V_\eta \to F^d$, and denote their joint map by $\pi : W \to K^d$. Solving the defining linear equations expresses the unique element of V_η with $\pi(T) = c$ rationally in the kernel coordinates and linearly in c. The condition that this element belongs to U_F defines a nonempty open subset of F^d. Since K is infinite, K^d is Zariski dense in F^d; choose $c \in K^d$ in that open subset.

After restricting to a nonempty open subset of $\overline{\mathcal{M}}$, the subspaces V_L have constant dimension, $\pi|_{V_L}$ is invertible, and the solutions with $\pi(T) = c$ belong to U. Every fiber there consequently meets the affine slice $S = \pi^{-1}(c)$ in exactly one point, varying rationally with L. These solutions and $\Psi|_S$ are inverse on nonempty open subsets. As $S \cong \mathbb{A}_K^{\dim W - d}$, the closure is rational. □

For $V = K^3$, a plane is the kernel of a nonzero linear form (a, b, c), unique up to a nonzero scalar. Thus

$$\mathrm{Gr}(2, K^3) \xrightarrow{\sim} \mathbb{P}^2_K, \qquad \ker(a, b, c) \longmapsto [a : b : c],$$

where $[a : b : c]$ denotes the corresponding homogeneous coordinates.

Proposition A.2. *For shape $(3, 3)$, the generic normalized kernel dimensions are $(\ell_0, \ell_1, \ell_2) = (2, 0, 0)$, and the kernel image is*

$$\mathcal{M} \cong \mathbb{P}^2_K \setminus \{[1 : 1 : 1]\}.$$

For every $T \in U$,

$$H_1(\langle T \rangle) \cong K, \qquad H_j(\langle T \rangle) = 0 \quad (j \neq 1).$$

Proof. Write $T = (t_{ij})$ and $(a, b, c) = (t_{00}, t_{11}, t_{22})$. In degree zero, f_T is the row $(a, b, c) : K^3 \to K$. In degree one, the columns corresponding to the three constant maps $[1] \to [2]$ are aJ, bJ, cJ, where J is the all-ones 2×2 matrix. When $(a, b, c) \neq 0$, they span KJ. Modulo this line, the other three columns are represented by

$$B_T = \begin{pmatrix} t_{01} - a & t_{02} - a & t_{12} - b \\ t_{10} - a & t_{20} - a & t_{21} - b \\ b - a & c - a & c - b \end{pmatrix}.$$

Its first two rows are arbitrary, while its last row is nonzero exactly when a, b, c are not all equal. Consequently $f_{T,1}$ can have rank 4 for a specified diagonal precisely when that diagonal is non-constant. Such a choice gives $\operatorname{rank} g_{T,0} = 1$ and $\operatorname{rank} g_{T,1} = 3$, using the natural decomposition $P_1 = C_1 \oplus s_0 C_0$ and its counterpart for X.

The normalized simplex complex $C = N(K[\Delta^2])$ has dimensions $(3, 3, 1)$ and injective differential $C_2 \to C_1$. Since $g_{T,1}$ is injective, the chain-map identity makes $g_{T,2}$ injective as well. These

are maximal ranks in all degrees, and $U = \{T : \det B_T \neq 0\}$. Thus on U the kernel complex is $\ker(a, b, c)$ in degree zero and zero elsewhere. Conversely, every nonconstant projective diagonal is realized by choosing the first two rows of the displayed matrix to complete its last row to a basis. This construction works over every residue field. Under the displayed identification of $\mathrm{Gr}(2, K^3)$ with $\mathbb{P}^2_K$, the kernel map therefore has exactly the stated image.

The restriction of the augmentation $(1, 1, 1)$ to $\ker(a, b, c)$ is nonzero, because (a, b, c) is not proportional to $(1, 1, 1)$. It is therefore surjective with one-dimensional kernel. Formula (8) proves the homology assertions. □

For comparison, the all-ones 3×3 tensor generates the constant simplicial module K: every simplicial operator sends it to an all-ones tensor. Its homology is K in degree zero and zero in positive degrees. Thus generated submodules distinguish tensors by homology even when their common ambient module is contractible.

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